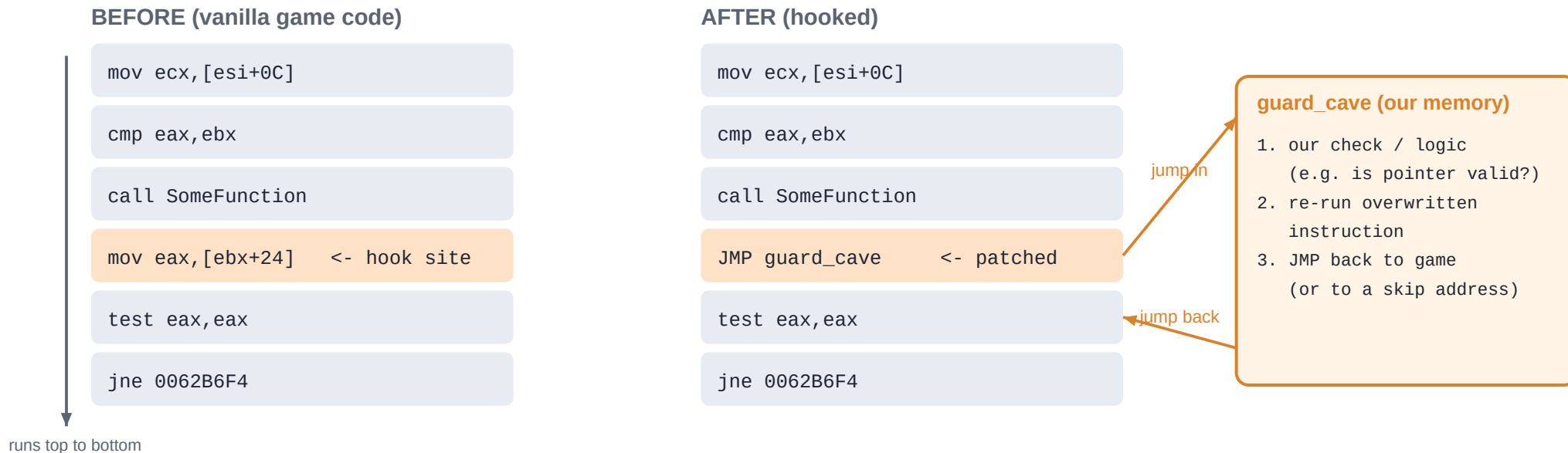


What is a code cave?

L.A. Noire Wanted System (wanted_lab_v2.lua) | LAnoire.exe base 004A0000 | 27 caves installed

A code cave is a small block of our own machine code placed in free memory inside the game process. To use it, we overwrite 5+ bytes of the game's code at a chosen address (the hook) with a JMP to the cave. The cave runs our logic, re-runs the game instructions we overwrote, then Jumps back so the game carries on. Because the cave runs on the game's own thread, it can safely call game functions (spawn, give weapon, re-plan).



- | | |
|-------------------------------------|--|
| <div></div> Workers | Run our jobs on the game thread each frame (handles, re-plans, spawns, gives). |
| <div></div> Trackers / Audio | Observe or tweak one small behaviour (aim target, mute fake shots). |
| <div></div> Teardown hook | Cleans our data when the game recycles an actor. |
| <div></div> Crash guards | Check a pointer before the game uses it; skip the code if it looks invalid. |
| <div></div> Logger / shield | Diagnostics (guard20) and a stack-alignment workaround (guard22). |

Static address = dynamic address - 0xA0000 (the exe loads at 004A0000 instead of its preferred 00400000). Code addresses are stable across sessions.

Where the caves sit in the game

Grouped by the game system each hook patches. Colour = category.

Wanted-system core

mw_cave	0x005CB280
System Worker	
ps_cave	0x0090A420
Spawner Worker	
tp_cave	0x006262B0
Teardown Hook	
wa_cave	0x005A38D0
Aim Tracker	
silence_cave	0x005A5B44
Audio Controller	

Perception / awareness

guard7_cave	0x0062BD23
Crash Guard	
guard12_cave	0x0062B6D6
Crash Guard	
guard13_cave	0x0062B690
Crash Guard	
guard22_cave	0x0062C0D5
Vector Shield	
guard15_cave	0x006091F0
Crash Guard	
guard8_cave	0x00AC8F45
Crash Guard	

Relationship manager

guard5_cave	0x00C18934
Crash Guard	
guard6_cave	0x00C17CBC
Crash Guard	
guard9_cave	0x00C18918
Crash Guard	
guard21_cave	0x00CDB3CA
Crash Guard	

Squads & tactics

guard10_cave	0x00790E38
Crash Guard	
guard19_cave	0x00790F68
Crash Guard	
guard14_cave	0x008DDAA0
Crash Guard	

Cover system

guard1_cave	0x0062DAFB
Crash Guard	
guard2_cave	0x0062DA2D
Crash Guard	
guard4_cave	0x0077E344
Crash Guard	
guard4b_cave	0x0077E377
Crash Guard	

Behaviour / AI components

guard3_cave	0x007251AD
Crash Guard	
guard16_cave	0x00725B90
Crash Guard	

Interesting events

guard17_cave	0x00ACC5DC
Crash Guard	
guard18_cave	0x00ACCA92
Crash Guard	
guard20_cave	0x00ACCDAA
Diag Logger / Guard	

Count by category

Crash Guard	20
System Worker	1
Aim Tracker	1
Teardown Hook	1
Spawner Worker	1
Audio Controller	1
Diag Logger / Guard	1
Vector Shield	1

Total: 27 caves (5 functional, 22 guards/shields)

Full cave reference

Part 1 of 2 (from the agent's cave list)

Cave	Hook (dynamic)	Static	Category	Purpose & what it does
mw_cave	0x005CB280	0x0052B280	System Worker	Main thread tick hook. Manages cop weak-ref handle allocation (initial rc=1), re-planning behavior trees, and handle release queues on the game engine thread.
wa_cave	0x005A38D0	0x005038D0	Aim Tracker	Aim marker hook. When Cole is aiming (return address 0x005A7BB3), stores the target ActorInstance for aim awareness without altering combat logic.
tp_cave	0x006262B0	0x005862B0	Teardown Hook	Actor pool return hook. Intercepts actor culling on the game thread to strip all relationship pairs before the actor's handle is recycled.
ps_cave	0x0090A420	0x0086A420	Spawner Worker	Population-update hook. Executes queued GiveWeapon calls, AI template spawns, and AddAwareness events on the game thread.
silence_cave	0x005A5B44	0x00505B44	Audio Controller	Sound controller hook. Mutes weapon sound and muzzle flash when emitting fake gunshot stimuli for hostile pulses.
guard1_cave	0x0062DAFB	0x0058DAFB	Crash Guard	Weapon holster check guard. Checks if weapon controller pointer is null before checking draw state to avoid AV crash.
guard2_cave	0x0062DA2D	0x0058DA2D	Crash Guard	Dynamic cover candidate guard. Validates candidate actor pointer before reading cover offset [+0x0C].
guard3_cave	0x007251AD	0x006851AD	Crash Guard	Behavior tree evaluator guard. Validates behavior data pointer and vtable before calling behavior update [+0x20].
guard4_cave	0x0077E344	0x006DE344	Crash Guard	Cover point search guard. Validates cover list pointer and array bounds before iterating candidate cover spots.
guard4b_cave	0x0077E377	0x006DE377	Crash Guard	Cover slot lookup guard. Validates indexed cover slot pointer [edx+ebx*4] before reading cover reservation flag.
guard5_cave	0x00C18934	0x00B78934	Crash Guard	Relationship pair bounds guard. Validates relationship array bounds and table head pointer [+0x1B4].
guard6_cave	0x00C17CBC	0x00B77CBC	Crash Guard	Relationship target vtable guard. Validates target actor vtable pointer before calling virtual relationship method [+0x24].
guard7_cave	0x0062BD23	0x0058BD23	Crash Guard	Perception table iteration guard. Validates perception candidate pointer before reading actor entry [+0x1F4].

Full cave reference

Part 2 of 2

Cave	Hook (dynamic)	Static	Category	Purpose & what it does
guard8_cave	0x00AC8F45	0x00A28F45	Crash Guard	Target evaluation float guard. Checks if candidate actor is null before loading float coordinate [+0x9E0] into XMM0.
guard9_cave	0x00C18918	0x00B78918	Crash Guard	Relationship source vtable guard. Validates dead/recycled cop vtable before calling virtual relationship method [+0x20].
guard10_cave	0x00790E38	0x006F0E38	Crash Guard	Squad tactics loop 1 guard. Validates dead squad member vtable before calling virtual tactics method [+0x24].
guard12_cave	0x0062B6D6	0x0058B6D6	Crash Guard	Combat awareness flag guard. Validates candidate actor pointer before reading dead flag byte [+0xC29].
guard13_cave	0x0062B690	0x0058B690	Crash Guard	Awareness array iteration guard. Validates awareness list base pointer before indexed read [eax+ecx*4].
guard14_cave	0x008DDAA0	0x0083DAA0	Crash Guard	Squad scanner type guard. Validates object vtable before calling squad method [+0x24] on non-actor recycled handles.
guard15_cave	0x006091F0	0x005691F0	Crash Guard	Target evaluation safety guard. Validates target actor and AI component pointer [+0xC8] on dead/falling victims.
guard16_cave	0x00725B90	0x00685B90	Crash Guard	AI component candidate guard. Validates component pointer before reading component field [+0x94].
guard17_cave	0x00ACC5DC	0x00A2C5DC	Crash Guard	Interesting event manager guard. Validates callback object vtable before virtual method call [+0x28].
guard18_cave	0x00ACCA92	0x00A2CA92	Crash Guard	Event manager actor update guard. Validates event callback object before virtual method call [+0x28].
guard19_cave	0x00790F68	0x006F0F68	Crash Guard	Squad tactics loop 2 guard. Validates member actor and vtable before calling virtual tactics method [+0x24].
guard20_cave	0x00ACCDAA0	0x00A2CDA0	Diag Logger / Guard	CoverManager/InterestingActor hook. Validates actor pointer and logs actor pointer, caller return address, and pre-clear handle to clear_log.
guard21_cave	0x00CDB3CA	0x00C3B3CA	Crash Guard	ActorInstance freed pointer guard. Validates instance pointer before reading profession/type byte [+0x73].
guard22_cave	0x0062C0D5	0x0058C0D5	Vector Shield	Perception stack alignment shield. Traps 16-byte misalignment on [esp+0xA4] and loads centroid safely using movups.